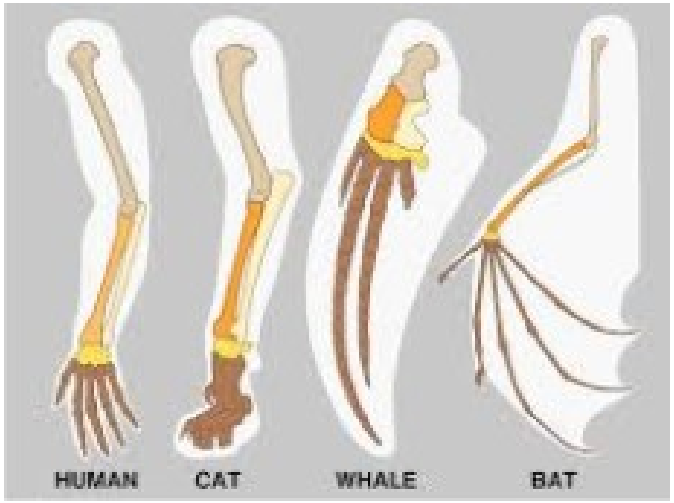
Evidence for Evolution



Comparative Studies in Anatomy

Embryology

Homologous Structures

Vestigial Structures

Geographical Distribution

Comparative Anatomy - Intro

- Compares structural features of organisms, looking at degree of similarity and therefore relatedness.
- 3 areas looked at:

  - Embryology: comparing early embryonic development
  - Homologous organs: organs that are similar in structure but may be used/adapted for different conditions
  - Vestigial organs: organs that may once have been important by have lost function

Embryology

- Provides evidence for evolutionary change over time
- Compares early development
  - Embryos initially very similar between species
  - Differentiation occurs during development
  - Differences become apparent later with more closely related organisms.

Evidence for common ancestry

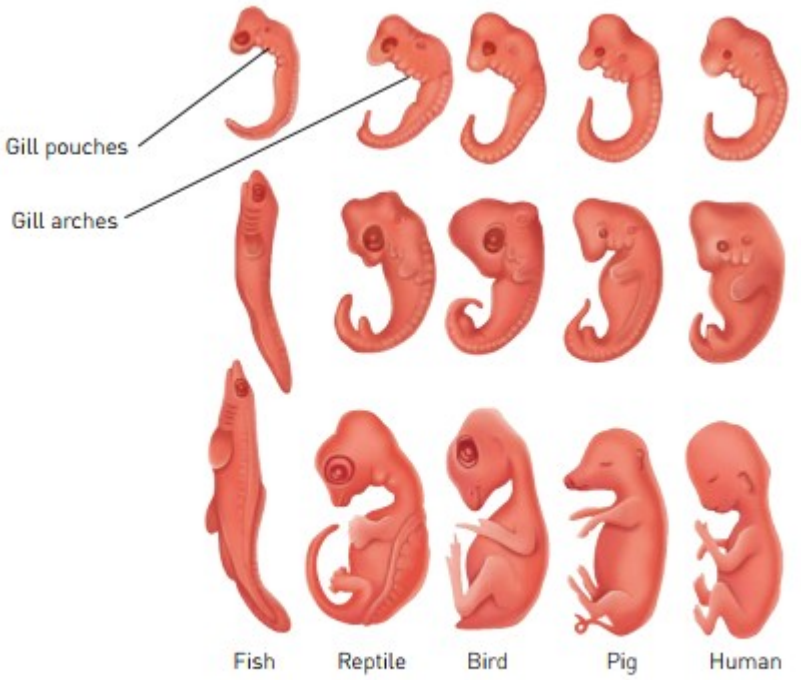


Figure 15.5 Embryos of five vertebrates at early (top), intermediate (middle) and late (bottom) stages of development

Note:

- All species start with similar features
- Gill arches appear in all – why, if mammals don't need them?
- In humans:
  - One gill slit becomes the Eustachian tube (connects ear to throat)
  - Tissue from around other gills develops into thryoid and tonsils

Homologous Structures

- Same bones appear in a variety of different species, but are used for different functions
- Illustrates common ancestry, then adaptation.

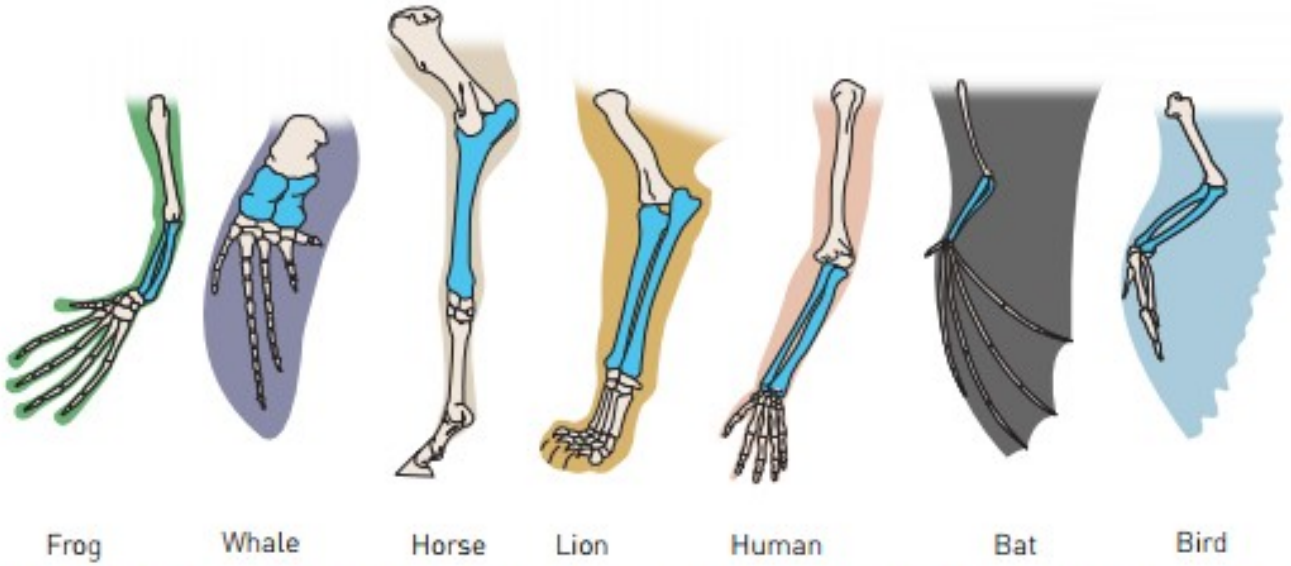


Figure 15.6 Left forelimb bones of seven vertebrates; from left to right: frog forelimb, whale flipper, horse forelimb, lion forelimb, human arm, bat wing, bird wing

- Example: vertebrate forelimbs
  - Same bones appear
  - Bones arranged in a similar way, but adapted in form to suit function
  - In pic below: forelimb bones

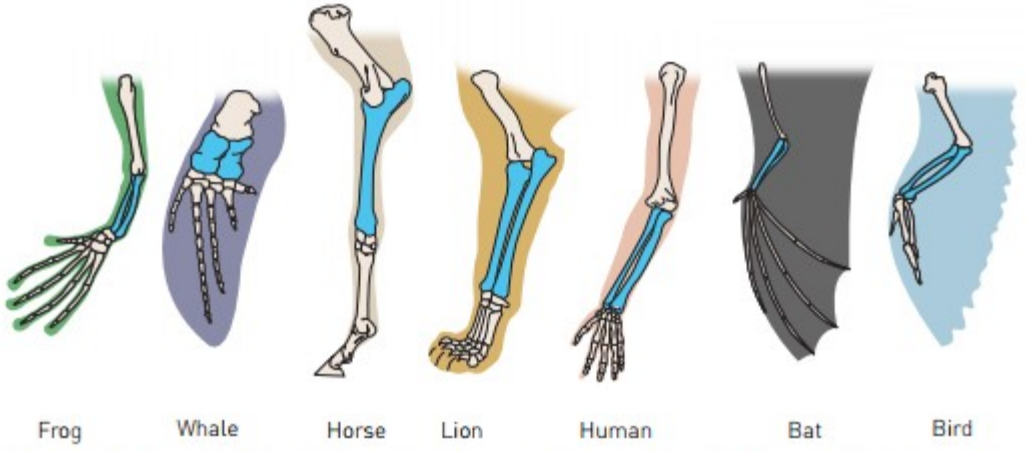


Figure 15.6 Left forelimb bones of seven vertebrates; from left to right: frog forelimb, whale flipper, horse forelimb, lion forelimb, human arm, bat wing, bird wing

- Example: Anthropoids (great apes) and Humans
- High degree of similarity, even across a range of habitats
  - Shows that evolutionary divergence is relatively recent

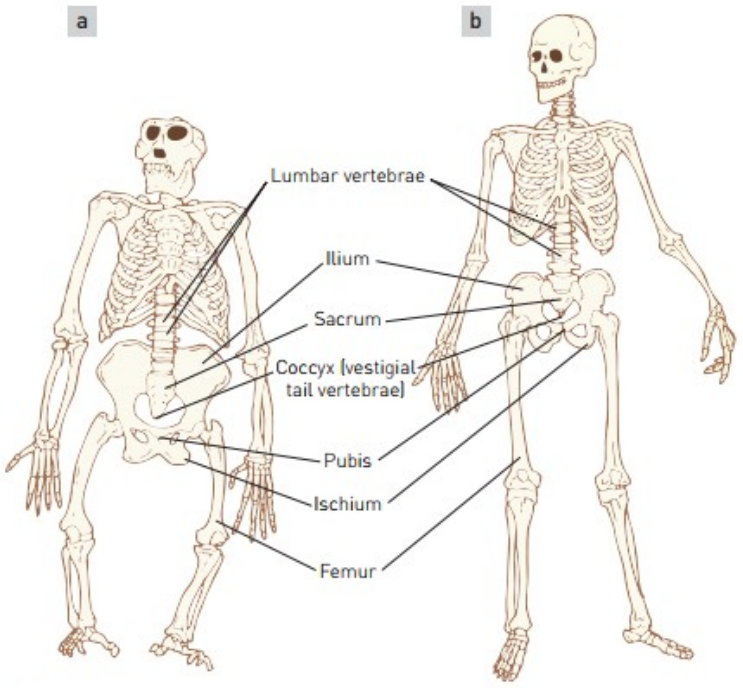


Figure 15.7 Skeleton of **a** a gorilla and **b** a human

Vestigial Organs

- Structures that have no or limited use in the organism, but were of use in an ancestral form.
- Evidence that the organism has evolved from an ancestor – the process isn't perfect!
- Eg:
  - Snakes sometimes born with vestigial limbs
  - Humans:
    - Nictitating membrane at corner of eye – used in cats, not used in humans
    - Wisdom teeth - some people don't have them!
    - appendix

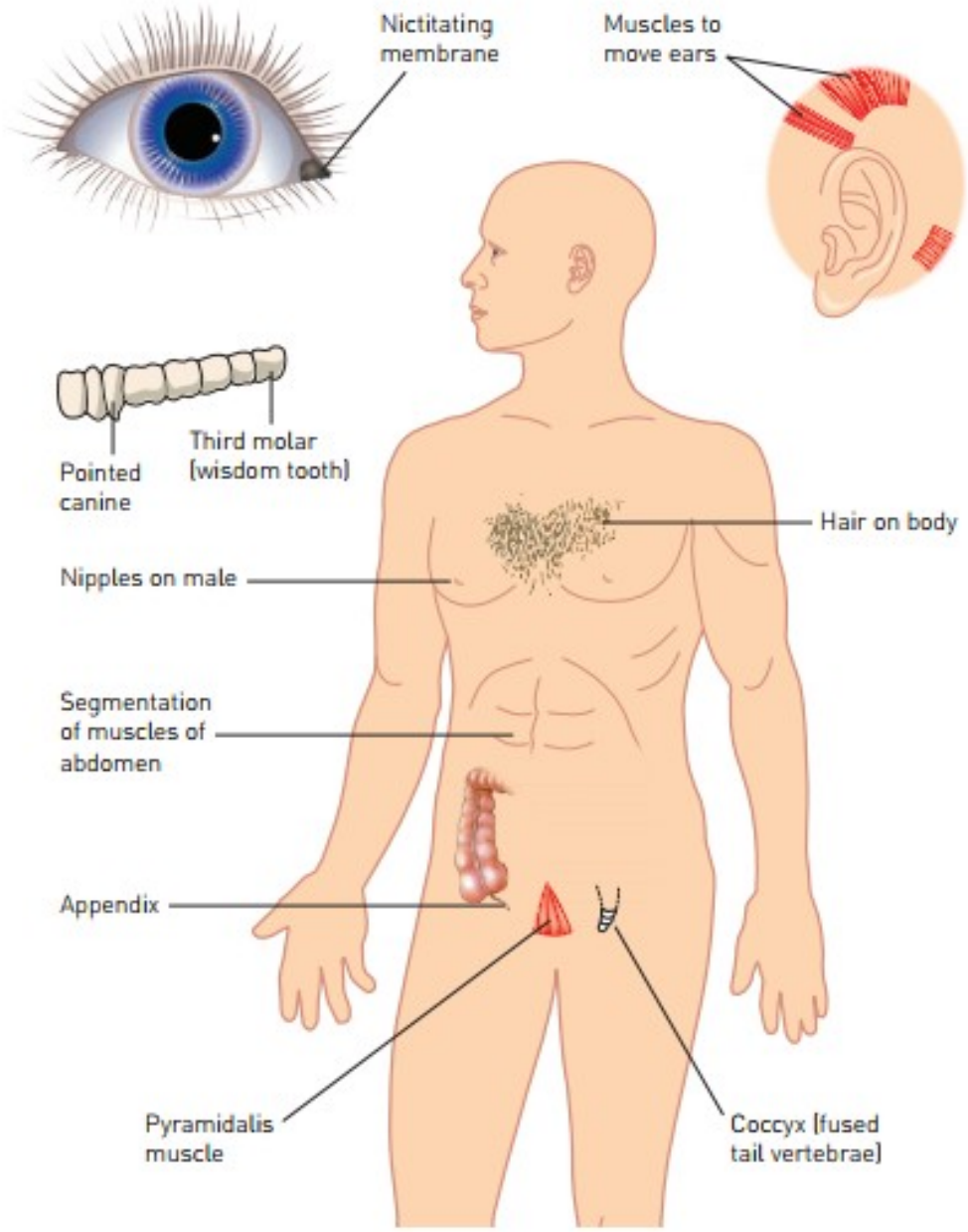


Figure 15.9 Some vestigial organs of humans

Geographical Distribution

- Further evidence for evolution
- Eg Darwin's finches – Galapagos Islands
  - Finches different in the islands than South American mainland
  - Several finches, back in time, had travelled from mainland, by chance – blown by a storm??
  - As a result, no competition , so evolved to take advantage of food sources on offer on different islands
    - Beaks adapted to different food sources.
  - Eventually evolved into 13 separate species on the islands.

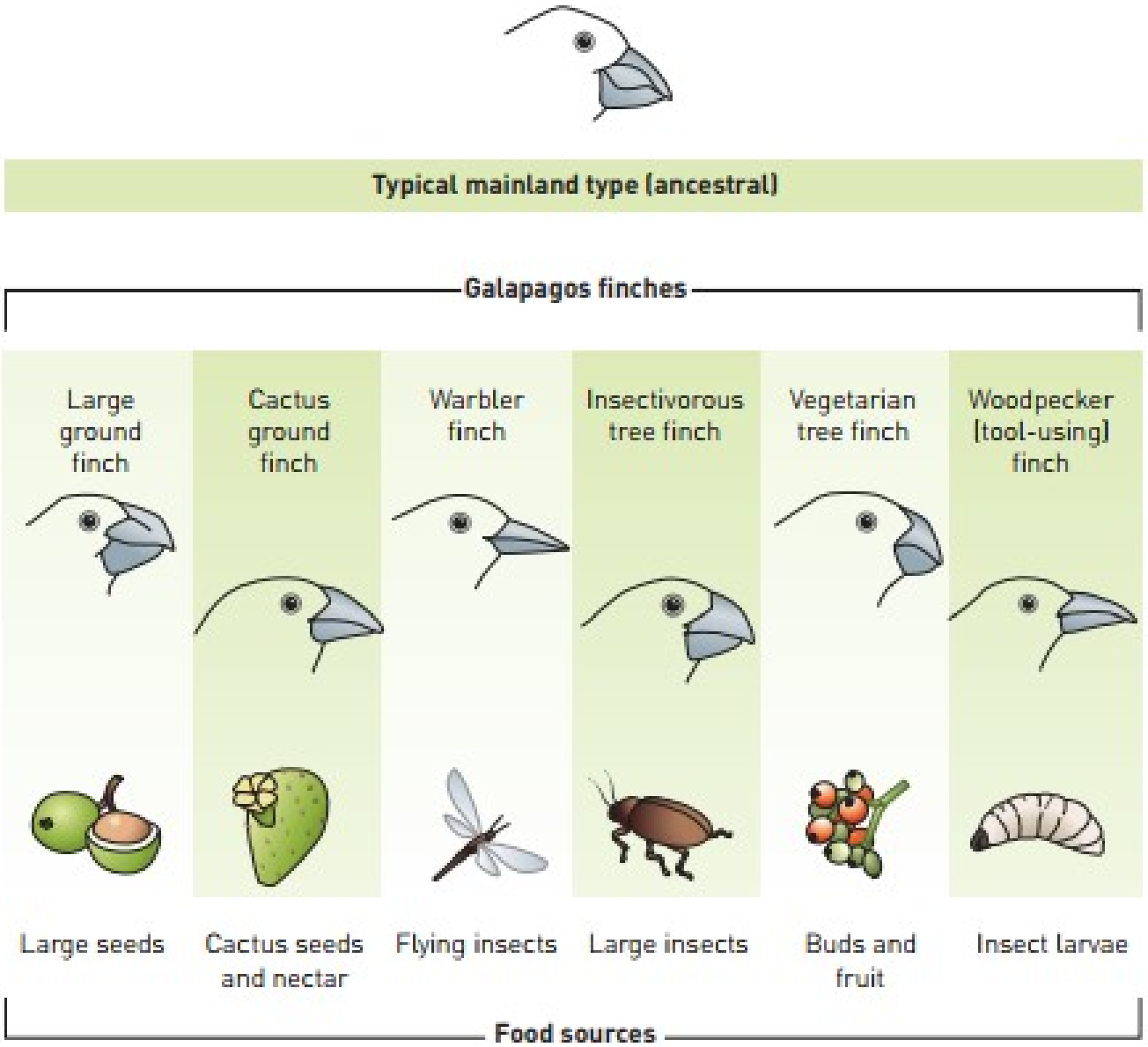


Figure 15.10 The finches of the Galapagos Islands. A common ancestor has evolved into many different species of finch as each adapted to a different food source.

Galapagos finches cont…

- Darwin hypothesised:
  - Mainland species arrived on islands and then diverged due to natural selection
  - First finches to arrive would have shown small, natural variations in beak shapes
    - Different finches would find different food sources easier to feed on – eg: one might find cactus seeds easiest to eat, another might find insect larvae easiest
    - Food source on individual islands would have favoured certain beak types over others, conferring a survival advantage
    - Birds on islands would slowly diverge in beak shape, adapting to food source on the islands.

Humans, Primates and evidence

- Humans most resemble chimpanzees and gorillas
- Therefore hypothesis that common ancestor most likely lived in place where chimps and gorillas are (Africa) rather than where lemurs or New World Monkeys are.
- This has been shown to be true – discovery of ancestral fossils in African Rift Valley.